



OVERCOMING CHALLENGES IN THE EYE CARE SERVICES

THE TECHNOLOGY-LED HEALTHCARE REVOLUTION

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Dr. E RAVINDRA MOHAN
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What do you feel are the major transformational changes in Indian healthcare today?

Developments in the field of Ophthalmology in our country, in recent times, can help highlight the transformation. The cutting edge of this transformation has been provided by institutions, innovation, indigenization, inexpensive solutions, and involvement of society and community, which one may call the 5 i's.

Can you elaborate on the role of institutions?

World class institutions in the country have provided a huge impetus to transforming eye care in India. These include Sankara Nethralaya, Aravind Eye Care System, L V Prasad Eye Institute, and RP Centre at AIIMS, New Delhi. In addition, many other excellent institutions have come up over time and this chain reaction has been truly transformative, pushing the pace of change.

In what ways have innovations contributed to this transformative process?

The most remarkable innovations have been in the delivery of eyecare services. From eye camps in rural areas, to well equipped vans taking diagnostic and curative facilities to remote areas, the base hospital approach for the performance of high-quality, high volume low-cost cataract surgery, the hub and spoke model for the delivery of specialized services are among the note-worthy ones. Innovations in surgical techniques like small incision non-phacoemulsifica-

tion cataract surgery have also truly revolutionized cataract surgery.

What do you mean by indigenization? How has this helped?

In a technology intensive field like ophthalmology, the cost of acquiring and maintaining cutting edge bio-medical equipment, instruments and implants pose a real challenge to the penetration of services and their utilization. This realization has stimulated technology research, indigenization, and the manufacture of slit lamps, operating microscopes, phacoemulsification machines for cataract surgery and a host of other items needed for state-of art eye care in our country. Made in India medical devices and surgical equipment are being exported, particularly to the developing world.

Your description of inexpensive solutions as a major transformative factor is an interesting one. Can you elaborate?

Cost is a reality, and a deciding one for that. A combination of innovative solutions, indigenization of manufacturing, price controls, economies of scale and healthy competition among the players has ensured costs lower than almost anywhere in the world. Good quality intra-ocular lenses, implants used in cataract surgery in millions of operations performed every year, are available to eye care institutions at as low a price as rupees 70 or so apiece, which in international

terms, is less than a dollar. The low cost of this vital implant helps keep the price of high quality high-volume cataract surgery at a very reasonable level.

What do you mean by the involvement of society and community?

Eye care touches lives in a very intimate way with an estimated 70 % of the population needing eye care services. From embracing services offered, to driving the need for better services as in the case of stitchless cataract surgery, to active support by local philanthropists and through corporate social responsibility initiatives, eye care services and initiatives have received ample support.

What are the exciting technologies in evolution that you see revolutionizing healthcare in the coming few years?

Most of the disruptions and revolutions in healthcare are going to be in the digital space with artificial intelligence expected to play an important role. Digitization of medical records permitting near instantaneous access to patient and healthcare providers is a likely game changer. The use of tele-medicine tools and platforms are expected to literally take health care to the doorstep of those in need. Advanced treatment algorithms, data analytics, and robotic surgical systems are likely to improve treatment planning, customize care, and lead to vastly improved outcomes of treatment.